The CSP Dictionary

A CIF-based data standard for crystal structure prediction methods

The CSP DS Team

Nov 2026

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advancing structural science

Agenda

- Introduction and summary of the project
- Updates and new features of the dictionary
  - New data block classes (structure similarity and conformers generation)
  - Structure properties
  - Cross references
  - Space Search variables and constraints
- Discussion
- Publication draft

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The CSP Data Standards Initiative

What is it?

A community initiative that aims at developing a common language for describing CSP studies and their generated structures.

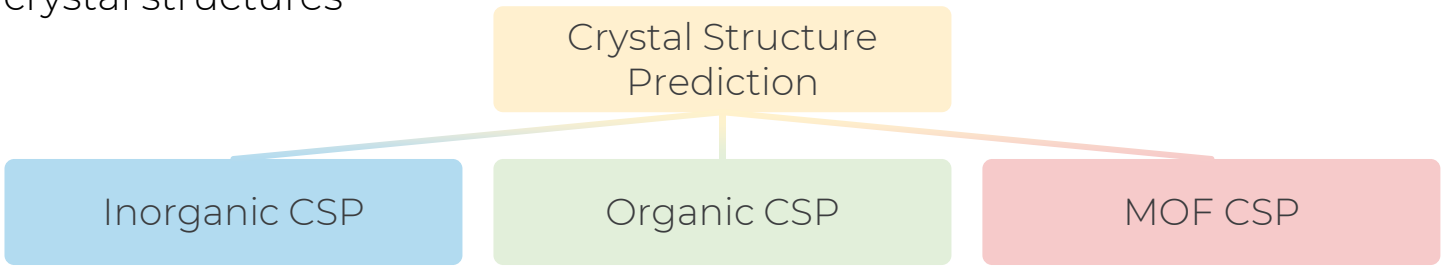
Why does it help?

- CSP studies are often complex and can produce large amounts of data.
- Different codes and communities may record CSP studies using different formats, terminology, or levels of detail.
- This makes it harder to compare results, reproduce studies, build shared benchmarks or databases, and apply post-analysis tools.
- A standard helps by defining how CSP studies should be described in a consistent and reusable way.

The CSP Data Standards Initiative



- An inclusive initiative that establishes **community standards** for predicted crystal structures

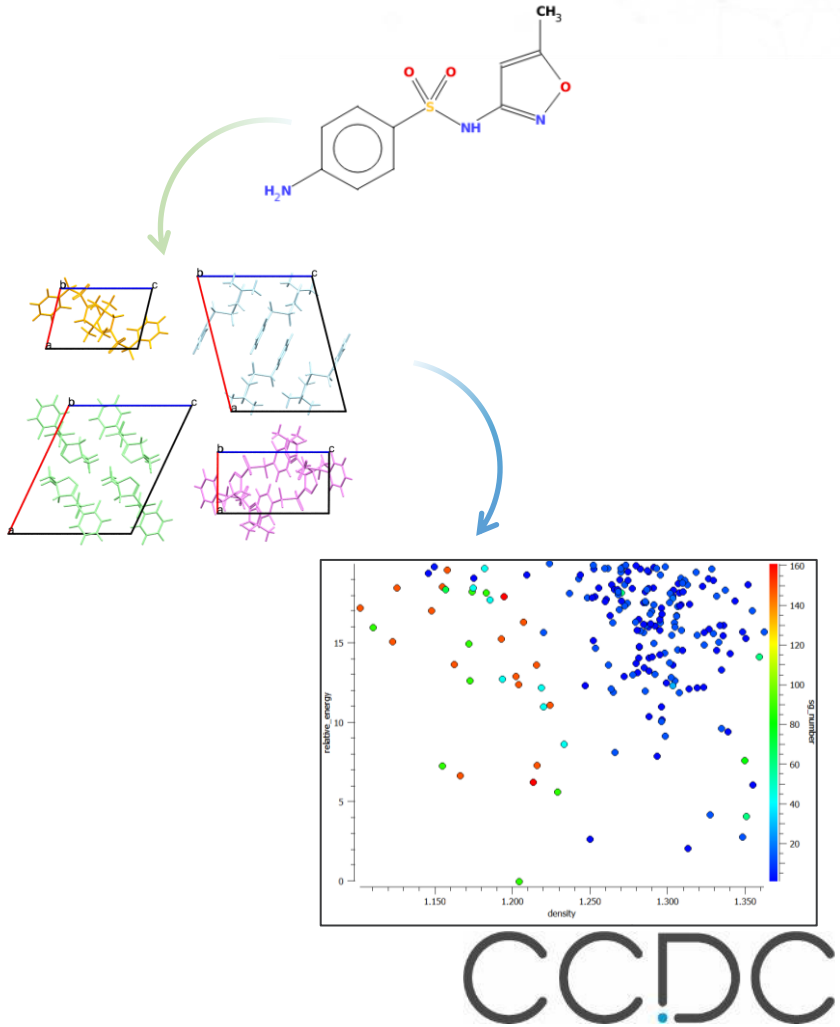


- Supports level of **detail** deemed appropriate by different communities
- **Flexible structure** for future expansion as CSP methods are continuing to evolve
- The dictionary is based on the DDLm format (CIF) and is designed to be **machine-readable**
- The focus of the project is on the development of a CSP dictionary describing structure generation methods and structure ranking methods

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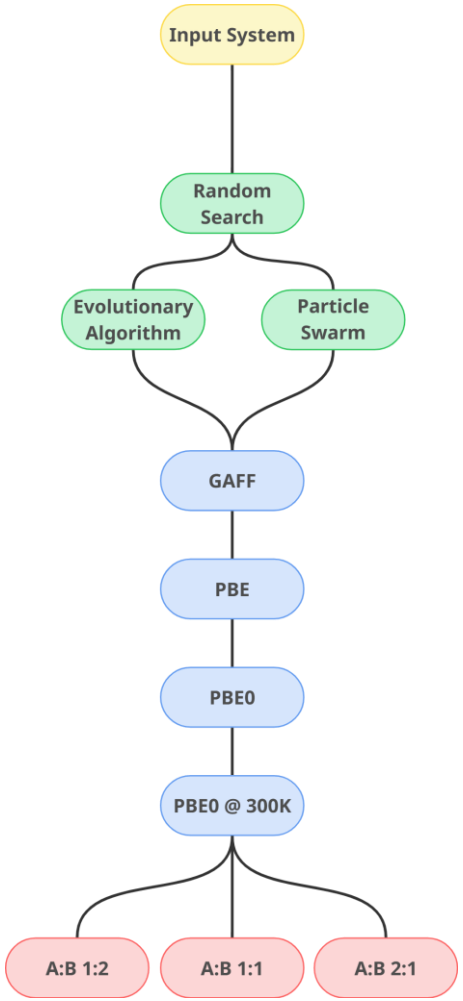
Crystal Structure Prediction

- CSP starts from a molecular structure or set of atomic species and searches for plausible crystal packings.
- These structures are then compared in terms of relative stability, often using lattice energy or free-energy-based ranking.
- The resulting crystal energy landscape is often presented as density vs energy plot.



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CSP Workflows



Multiple Generation methods

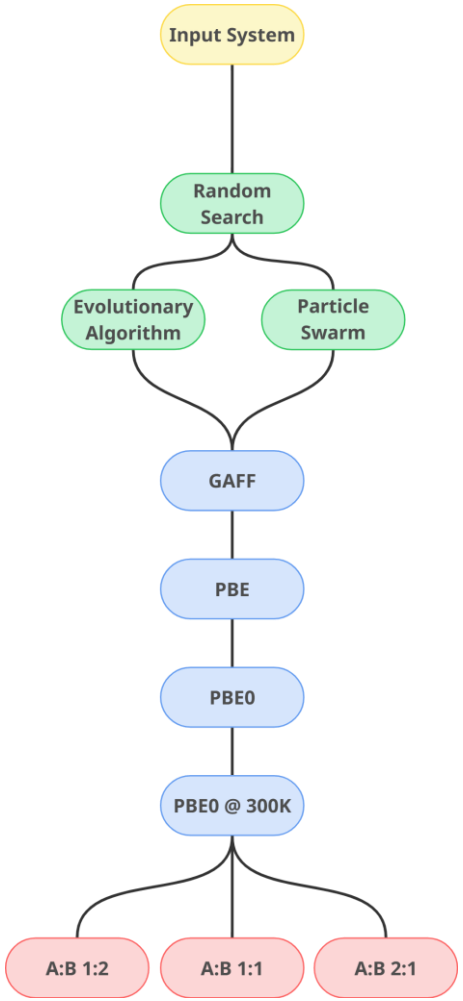
Multi-step Optimisation and Ranking

Multiple Stoichiometries

- A CSP dictionary should be able to accurately describe the complex workflows used in CSP today
- The same class of methods can be used more than once with different settings
- Inside a single CIF data block we cannot have duplicate data fields.

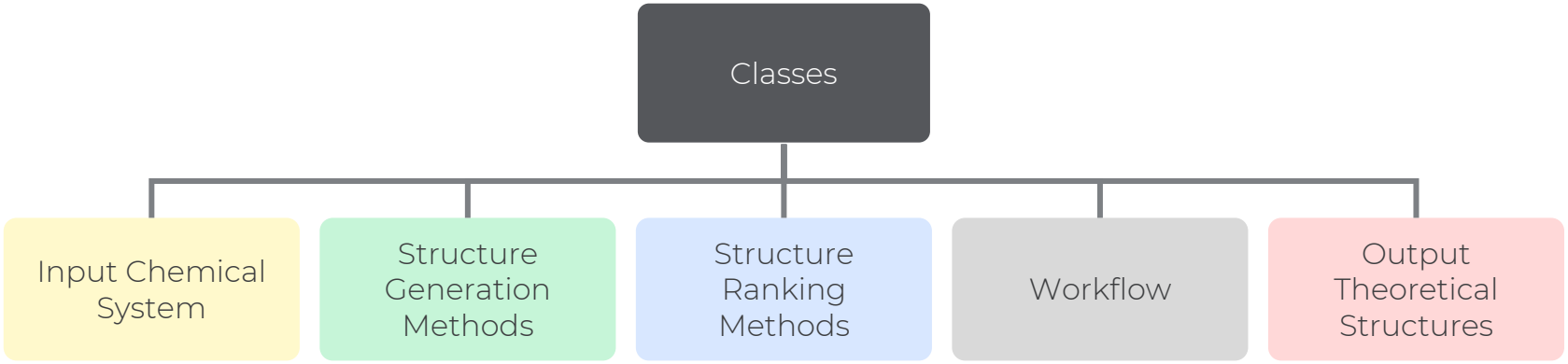
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Organisation into Data Blocks



Each data block groups together all the data fields describing a single computational structure-prediction step or output structure, keeping parameters, metadata, and resulting properties organised in a clear, machine-readable way.

The associated classes specify the type of information contained in each data block, making it easy to understand what aspect of the CSP workflow the block represents.



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Organisation into Data Blocks

Each part of a CSP calculation is stored in its own data block.

Essential fields in every data block:

- **Class**: Type of data block
  - Input
  - Generation Method
  - Ranking Method
  - Workflow
  - Theoretical Structure
  - Step
  - General Output
- **ID**: Unique identifier (typically a UUID) used for linking across files.
- (Optional) A human-readable **description** that provides a concise label to make files understandable to users.

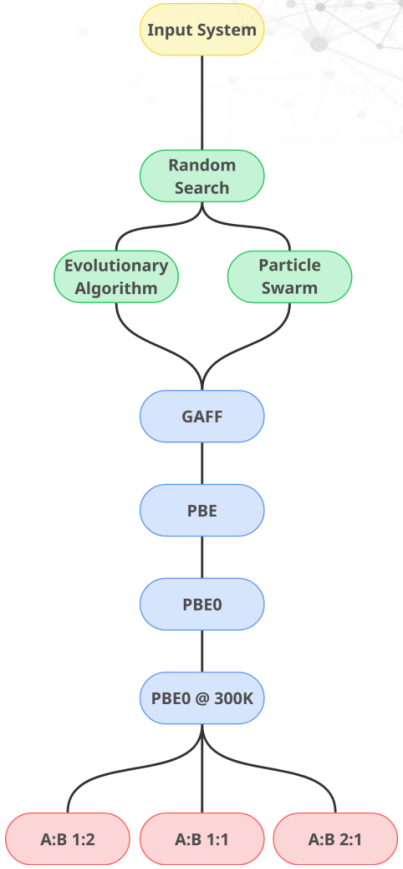
Input

```
data_cbn_tmp
# Datablock Details
_csp_data_block.class          "Input"
_csp_data_block.id             2a2611e3-2021-4b03-a7c6-0ef71239008f
_csp_data_block.description    “CBN-TMP”

_csp_input.name                “cannabinol_tetramethylpyrazine”
_csp_input.identifier          “BT-XXX”

# Molecules
loop_
    _csp_input_molecular_entity.number
    _csp_input_molecular_entity.identifier
    _csp_input_molecular_entity.smiles
    _csp_input_molecular_entity.name_common
    1 CBN “CCCCCc1cc(O)c2c(OC(C)(C)c3ccc([CH6])cc23)c1” cannabinol
    2 TMP “Cc1nc(C)c(C)nc1C”                          tetramethylpyrazine

# Type of Composition Search
_csp_input.composition_calculation  "fixed"
_csp_input.composition_coefficients [[1 2] [1 1] [2 1]]
```



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Data Blocks: Workflow

The **Workflow** datablock defines how different structure generation and ranking methods are connected into a full CSP workflow.

- Specifies the order of generation and ranking steps.
- Connects each step to the specific method data block via its unique ID.
- Tracks how structures flow from one ranking step to the next.

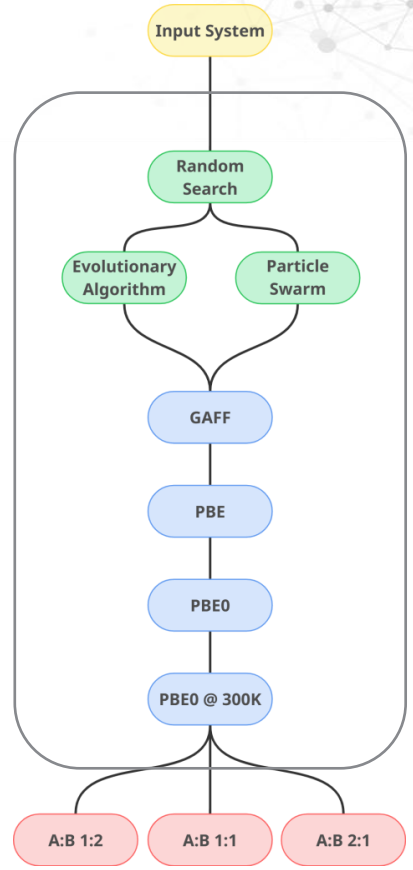
Additional data fields include descriptions and method names used for human-readability.

Workflow

```
data_workflow
    # Data blocks details
    _csp_data_block.class             Workflow
    _csp_data_block.description       csp_workflow_long
    _csp_data_block.id                425a77e3-a2a4-40fb-acdb-1e25b2d00bbd
    _csp_data_block.additional_files ["structure_generation_methods.cif",
                                      "structure_ranking_methods.cif"]

    # Structure Generation Methods
    loop_
        _csp_structure_generation.stage
        _csp_structure_generation.preceding_stage
        _csp_structure_generation.data_block_label
        _csp_structure_generation.data_block_description
        _csp_structure_generation.data_block_id
      0 .  "rs"    "Random Sampling"                1d5b2b53-d70d-44…
      1 0  "ea"    "Evolutionary Algorithm"         af534fed-8153-4a…
      2 0  "pso"   "Particle Swarm Optimisation"    242618a7-89b6-40…

    # Structure Ranking Methods
    loop_
        _csp_structure_ranking.stage
        _csp_structure_ranking.preceding_stage
        _csp_structure_ranking.data_block_label
        _csp_structure_ranking.data_block_description
        _csp_structure_ranking.relative_energy_cutoff
        _csp_structure_ranking.max_structures_retained
        _csp_structure_ranking.data_block_id
      0 .  "gaff"     "GAFF Optimisation"   30.0    20000  6f36e2d3-a1…
      1 0  "pbe"      "PBE Optimisation"    10.0     2000  97088479-72…
      2 1  "pbe0"     "PBE0 Single Point"   10.0       50  97773a57-f2…
      3 2  "pbe_qha"  "QHA"                    .        .  2b9deed1-11…
```



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The CSP and CompChem Dictionary

Input Chemical System

- Atoms
- Molecules
- Fixed or variable stoichiometry

Structure Generation Methods

- Evolutionary Algorithms
- Particle Swarm Optimisation
- MC Simulated Annealing
- MC Parallel Tempering
- Random Search

Structure Ranking Methods

- Periodic Density Functional Theory
- Forcefields
- Semi-Empirical
- Wavefunction
- Free Energy

Output Structures

- Energy and rank
- 3D Structure
- Pressure and temperature

CSP Dictionary

Computational Chemistry Dictionary

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The CSP and CompChem Dictionaries

Input Chemical System
- Atoms
- Molecules
- Fixed or variable stoichiometry
- *Search space, variables, constraints*
- *Landscapes Links*

Structure Generation Methods
- Evolutionary Algorithms
- Particle Swarm Optimisation
- MC Simulated Annealing
- MC Parallel Tempering
- Random Search

Intermediate Step
- *Structure Similarity*
- *Conformers Generation*
- *General Parameters*

General Output
- *Cumulative computational costs*
- *Single step global minima*

CSP Dictionary

Structure Ranking Methods
- Periodic Density Functional Theory
- Forcefields
- Semi-Empirical
- Wavefunction
- *ML Potentials*
- Free Energy

Output Structures
- Energy and rank
- 3D Structure
- Pressure and temperature
- *Structure Properties*
- *Structures Links*

Computational Chemistry Dictionary

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Data Blocks: General Outputs

The **Output** data block can be used to describe properties of the landscape, not specific to a single structure or of a single step.

- Links directly to the landscape input and workflow data block.
- Provides general information such as the hardware used to perform calculations
- Tracks down computational costs, structures analysed and global minima for each step

General Output

```
data_csp_output

# Datablock Details
_csp_data_block.class                 "Output"
_csp_data_block.description           csp_output
_csp_data_block.id                    1d327ab0-3f84
_csp_data_block.additional_files      ["csp_input.cif" "workflow.cif“ "structure.cif"]

# Input and workflow identifiers
_csp.output_input_system_id           2a2611e3-2021
_csp.output_workflow_id               964fcd15-82e1

# Output Details and Computational Cost
_csp.output_hardware_description      "AMD Zen 2 EPYC 7H12"
_csp.output_total_computational_cost  185101.0

loop_
    _csp_output.stage
    _csp_output.stage_id
    _csp_output.stage_computational_cost
    _csp_output.number_of_structures
    _csp_output.average_computational_cost
    _csp_output.global_minima
    _csp_output.global_minima_id
    # Generation Methods
    "rs"       6e0147be       1.0     7500  0.00013 . .
    "sa"       95f28b3c     100.0  2000000  0.00050 . .
    # Ranking Methods
    "gaff"     83f824d3  100000.0  2007500  0.04981 "structure_1_1"   63f70eef
    "psi_mol" d6f196c5   25000.0    50000  0.50000 "structure_105_2" 07457f98
    "pbe"      17ad684a   60000.0    10000  7.50000 "structure_105_3" 756c1329
```

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Data Blocks: Intermediate Steps

The **Step** data block provides flexible data fields to describe any intermediate step in a CSP workflow.

Intermediate Step

```
data_void_calc

# Datablock Details
_csp_data_block.class                "Step"
_csp_data_block.description          “Void Calculation”
_csp_data_block.id                   deb4fe89

# Method
_csp_step.method                     “Void”
_csp_step.method_description         “Void’s volume and type computation
_csp_step.software                   “CSD Python API”
_csp_step.software_version           3.7.0

# Parameters
loop_
    _csp_step_variable.label
    _csp_step_variable.variable
    “Probe Radius” 1.2
    “Grid Spacing” 0.3
```

Data Blocks: Intermediate Steps

The **Step** data block provides flexible data fields to describe any intermediate step in a CSP workflow.

Detailed data fields are given for **Structure Similarity** approaches and Conformers Generation.

3 types of structure similarity methods are supported:

- PXRD patterns
- Interatomic distance based approaches
- Molecular Shell approaches

These are useful for both duplicates' detection and experimental-theoretical structures similarity analysis

Intermediate Step

```
data_cps

# Datablock Details
_csp_data_block.class                                  "Step"
_csp_data_block.description                            “cps”
_csp_data_block.id                                     677de204

# Structure Similarity details
_csp_similarity.method                                 "Crystal Packing Similarity"
_csp_similarity.method_description                     "A molecular-shell-based algorithm"
_csp_similarity.software                               "CSD Python API"
_csp_similarity.software_version                       3.7
_csp_similarity.distance_cutoff                        1.0
_csp_similarity.energy_threshold                       0.5
_csp_similarity.density_threshold                      0.05
_csp_similarity.include_hydrogens                      False
_csp_similarity.optimise_cell                          False
_csp_similarity.molecular_shell_size                   30
_csp_similarity.molecular_shell_distance_tolerance     0.30
_csp_similarity.molecular_shell_angle_tolerance        30
```

Theoretical Structure

```
...
_theoretical_structure.experimental_match_id                ZEHFUR02
_theoretical_structure.experimental_match_database          CSD
_theoretical_structure.experimental_match_description       "Experimental Form B"
_theoretical_structure.experimental_match_similarity_method 677de204
_theoretical_structure.experimental_match_similarity_score  0.265
...
```

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Data Blocks: Intermediate Steps

The **Step** data block provides flexible data fields to describe any intermediate step in a CSP workflow.

Detailed data fields are given for Structure Similarity approaches and **Conformers Generation.**

New data fields can describe:

- Conformers' generation methods
- Geometry optimisation
- Clustering

Intermediate Step

```
data_cg

# Datablock Details
_csp_data_block.class                                "Step"
_csp_data_block.description                          cg
_csp_data_block.id                                   b7dcc1d4

# Conformers Generation
_csp_conformer.generation_method                     ConformerGenerator
_csp_conformer.generation_method_description         "Knowledge-based conformer generation"
_csp_conformer.generation_software                   "CSD Python API"
_csp_conformer.generation_software_version           3.7

# Optimisation
_csp_conformer.optimisation_method                   “PBE/cc-pvdz”
_csp_conformer.optimisation_method_description       "GGA DFT optimisation"
_csp_conformer.optimisation_software                 Psi4
_csp_conformer.optimisation_software_version         9.0
_csp_conformer.energy_cutoff                         20.0
_csp_conformer.max_number_of_conformers              50

# Clustering
_csp_conformer.clustering_method                     “MoleculeOverlay”
_csp_conformer.clustering_method_description         "Overlays two molecules"
_csp_conformer.clustering_software                   "CSD Python API"
_csp_conformer.clustering_software_version           3.7
_csp_conformer.clustering_rmsd_cutoff                0.4
```

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Structure Properties

- We included new energy data fields for phase diagrams and uncertainty.
- Per-atom forces and stress tensor.
- General material properties:

Electronic	Magnetic	Mechanical	Transport properties
• Band Gap • Partial charges	• Magnetic Ordering • Total Magnetic Moment	• Bulk Modulus • Elastic constants	• Seebeck coefficient • Carrier mobility

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Cross References

We introduced a series of data fields to specify custom relationships between *landscapes* and between *structures*

Input

```
...
loop_
    _csp_landscape_cross_reference.relationship_description
    _csp_landscape_cross_reference.related_id
    _csp_landscape_cross_reference.related_description
    _csp_landscape_cross_reference.related_location
    "Target ACE2-Lys353" 079b5459-07b4 "Mol1" "Internal DB"
    "Target ACE2-Lys353" 6cfd0766-eb75 "Mol2" "Internal DB"
    "Target ACE2-Lys353" f6442576-39ff "Mol3" "Internal DB"
    "Target ACE2-Lys353" d3aa26d3-cf6b "Mol4" "Internal DB"
...
```

Linked Landscapes

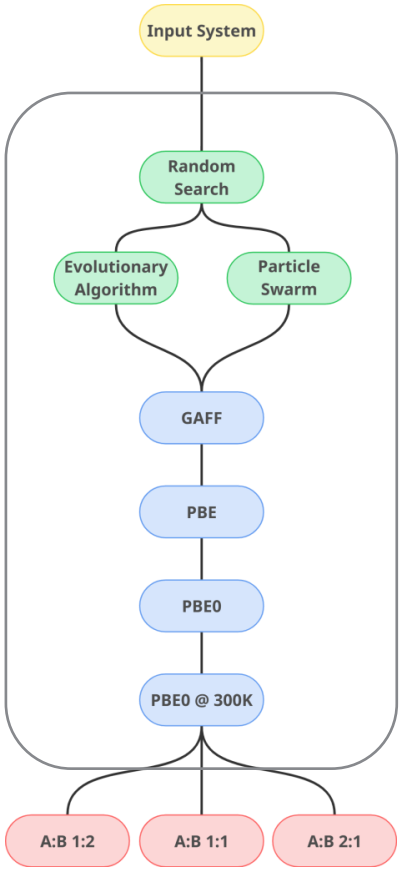
Theoretical Structure

```
...
loop_
    _theoretical_structure_cross_reference.relationship_description
    _theoretical_structure_cross_reference.related_structure_id
    _theoretical_structure_cross_reference.related_structure_description
    _theoretical_structure_cross_reference.related_structure_location
    _theoretical_structure_cross_reference.assessment_method
    _theoretical_structure_cross_reference.score
    _theoretical_structure_cross_reference.units
    "minor disorder component“ "BOVDUM" "Anhydrous Form I" "CSD"         "RMSD30" 0.345 "angstroms"
    "isostructural"            "MH-1"   "Monohydrate Form" "Internal DB" "RMSD30" 0.798 "angstroms"
...
```

Linked Structures

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Search Space



The *workflow*, *generation* and *ranking* stages, with any intermediate *step*, specify the **general CSP protocol**.

The data fields in these classes should be as independent as possible from the systems studied.

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Search Space

Generation Method

```
data_pso

# Datablock Details
_csp_data_block.class       "Generation Method"
_csp_data_block.id          242618a7-89b6-40d1-a6c4-cc51fd107a13
_csp_data_block.description "pso"

# Method Details
_csp.structure_generation_method      "Particle Swarm Optimisation“

_csp_particle_swarm_optimisation.population_size       50
_csp_particle_swarm_optimisation.inertia_weight        0.7
_csp_particle_swarm_optimisation.max_inertia_weight    0.9
_csp_particle_swarm_optimisation.min_inertia_weight    0.4
_csp_particle_swarm_optimisation.cognitive_coefficient 2
_csp_particle_swarm_optimisation.social_coefficient    2

# Search Space & Stopping Criteria
_csp.structure_generation_space_group_number_list [14 2 15 61 19 4 33 29 5 1]
_csp_structure_generation_stopping_criteria.description  "Max Structures"
loop_
    _csp_structure_generation_search_space.z_prime
    _csp_structure_generation_search_space.space_group_number_list
    _csp_structure_generation_stopping_criteria.max_structures_evaluated
    1 [14 2 15 61 19]  50000
    2 [14 2 15 61 19]  25000
    1 [4 33 29 5 1]    25000
```

Input

```
data_molecule

# Datablock Details
_csp_data_block.class         "Input"
_csp_data_block.id            2a2611e3-2021
_csp_data_block.description   “URE_H2O“

_csp_input.name               "Urea Hydrate"

...

_csp_search.space_group_number_list        [14 2 15 61 19 4 33 29 5 1]
_csp_search.stopping_criteria_description  ["Max Structures"]

loop_
    _csp_stopping_criteria.z_prime
    _csp_stopping_criteria.space_group_number_list
    _csp_stopping_criteria.max_structures_evaluated
    [1] [14 2 15 61 19]  50000
    [2] [14 2 15 61 19]  25000
    [1] [4 33 29 5 1]    25000
```

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Search Space: Variables and Constraint

Input

```
...
# Search Variables
loop_
    _csp_search_variable.index
    _csp_search_variable.type
    _csp_search_variable.domain_type
    _csp_search_variable.target
    _csp_search_variable.variable
    _csp_search_variable.description
    1   cell_length continuous  .    a      "Cell Parameters"
    2   cell_length continuous  .    b      "Cell Parameters"
    3   cell_length continuous  .    c      "Cell Parameters"
    4   cell_angle  continuous  .    alpha  "Cell Parameters"
    5   cell_angle  continuous  .    beta   "Cell Parameters"
    6   cell_angle  continuous  .    gamma  "Cell Parameters"
    7   position    periodic    [1]  x1     "Fractional Coordinate"
    8   position    periodic    [1]  y1     "Fractional Coordinate"
    9   position    periodic    [1]  z1     "Fractional Coordinate"
    10  position    periodic    [2]  x2     "Fractional Coordinate"
    11  position    periodic    [2]  y2     "Fractional Coordinate"
    12  position    periodic    [2]  z2     "Fractional Coordinate"
    13  position    periodic    [3]  x3     "Fractional Coordinate"
    14  position    periodic    [3]  y3     "Fractional Coordinate"
    15  position    periodic    [3]  z3     "Fractional Coordinate"
    16  position    periodic    [4]  x4     "Fractional Coordinate"
    17  position    periodic    [4]  y4     "Fractional Coordinate"
    18  position    periodic    [4]  z4     "Fractional Coordinate"
```

Input

```
...
# Search Constraints
loop_
    _csp_search_constraint.index
    _csp_search_constraint.type
    _csp_search_constraint.target
    _csp_search_constraint.lower_bound
    _csp_search_constraint.upper_bound
    _csp_search_constraint.boundary_type
    _csp_search_constraint.allowed_values
    _csp_search_constraint.description
    1  variable              [1]          4.0     30.0  hard      .         .
    2  variable              [2]          4.0     30.0  hard      .         .
    3  variable              [3]          4.0     30.0  hard      .         .
    4  variable              [4]         60.0    120.0  hard      .         .
    5  variable              [5]         60.0    120.0  hard      .         .
    6  variable              [6]         60.0    120.0  hard      .         .
    7  variable              [7]          0.0      1.0  hard      .         .
    8  variable              [8]          0.0      1.0  hard      .         .
    9  variable              [9]          0.0      1.0  hard      .         .
    10 variable              [10]         0.0      1.0  hard      .         .
    11 variable              [11]         0.0      1.0  hard      .         .
    12 variable              [12]         0.0      1.0  hard      .         .
    13 variable              [13]         0.0      1.0  hard      .         .
    14 variable              [14]         0.0      1.0  hard      .         .
    15 variable              [15]         0.0      1.0  hard      .         .
    16 global                [density] 1200.0   1800.0  hard      .         .
    17 interatomic_distance  [3 4]        1.5    .       hard      .         "Si-O"
    18 interatomic_distance  [1 3]        2.1    .       hard      .         "Fe-Si"
    19 interatomic_distance  [1 4]        1.6    .       hard      .         "Fe-O"
    20 interatomic_distance  [2 3]        2.0    .       hard      .         "Mg-Si"
    21 interatomic_distance  [2 4]        1.5    .       hard      .         "Mg-O"
```

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Search Space: Variables and Constraint

Input

```
...
# Search Variables
loop_
    _csp_search_variable.index
    _csp_search_variable.type
    _csp_search_variable.domain_type
    _csp_search_variable.target
    _csp_search_variable.variable
    _csp_search_variable.description
    1  cell_length continuous  .          a        "Cell Parameters"
    2  cell_length continuous  .          b        "Cell Parameters"
    3  cell_length continuous  .          c        "Cell Parameters"
    4  cell_angle  continuous  .          alpha    "Cell Parameters"
    5  cell_angle  continuous  .          beta     "Cell Parameters"
    6  cell_angle  continuous  .          gamma    "Cell Parameters"
    7  position    periodic    [1]        x1       “COM Fractional Coordinate"
    8  position    periodic    [1]        y1       "COM Fractional Coordinate"
    9  position    periodic    [1]        z1       "COM Fractional Coordinate"
    10 position    periodic    [2]        x2       "COM Fractional Coordinate"
    11 position    periodic    [2]        y2       "COM Fractional Coordinate"
    12 position    periodic    [2]        z2       "COM Fractional Coordinate"
    13 rotation    periodic    [1]        theta11  “Rotation - C2 symmetry"
    14 rotation    periodic    [1]        theta12  "Rotation"
    15 rotation    periodic    [1]        theta13  "Rotation"
    16 rotation    periodic    [2]        theta21  “Rotation - C2 symmetry"
    17 rotation    periodic    [2]        theta22  "Rotation"
    18 rotation    periodic    [2]        theta23  "Rotation"
    19 torsion     periodic    [4 5 6 8]  phi1     "Torsion"
```

Input

```
...
# Search Constraints
loop_
    _csp_search_constraint.index
    _csp_search_constraint.type
    _csp_search_constraint.target
    _csp_search_constraint.lower_bound
    _csp_search_constraint.upper_bound
    _csp_search_constraint.boundary_type
    _csp_search_constraint.allowed_values
    _csp_search_constraint.description
    1   variable [1]    4.0     30.0  hard      .        .
    2   variable [2]    4.0     30.0  hard      .        .
    3   variable [3]    4.0     30.0  hard      .        .
    4   variable [4]   60.0    120.0  hard      .        .
    5   variable [5]   60.0    120.0  hard      .        .
    6   variable [6]   60.0    120.0  hard      .        .
    7   variable [7]    0.0      1.0  hard      .        .
    8   variable [8]    0.0      1.0  hard      .        .
    9   variable [9]    0.0      1.0  hard      .        .
    10  variable [10]   0.0      1.0  hard      .        .
    11  variable [11]   0.0      1.0  hard      .        .
    12  variable [12]   0.0      1.0  hard      .        .
    13  variable [13]   0.0    180.0  hard      .        "C2 axis"
    14  variable [14]   0.0    360.0  hard      .        .
    15  variable [15]   0.0    360.0  hard      .        .
    16  variable [16]   0.0    180.0  hard      .        "C2 axis"
    17  variable [17]   0.0    360.0  hard      .        .
    18  variable [18]   0.0    360.0  hard      .        .
    19  variable [19] 150.0    180.0  hard      .        "Planar or pyramidal configurations"
    20  global   [density] 1200.0   1800.0  harmonic  .        .
```

CCDC

The GitHub Repository

A public GitHub Repository is available at the QR on the left. In addition to dictionary files, the repository contains:

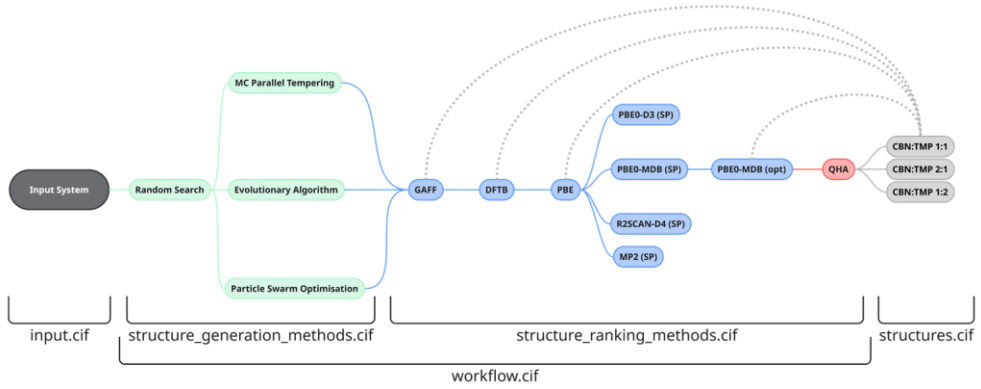
- Descriptive documentation of each new data field.
- Examples folder with a series of case studies of CSP protocols
- Slides from previous general meetings

New data fields can be proposed through issues and pull requests

1. Input Chemical System

This section specifies the atoms used in inorganic CSP or the input molecules for molecular crystal generation.

Category	Data Field	Type	Definition	Constraints	Units	Example
Input	`name`	char	See name_common and name_systematic from Core CIF dictionary.	Free Text		urea hydrate
Input	`composition_calculation`	char	"fixed" or "variable" composition calculation.	- Fixed - Variable		Fixed
Input	`composition_coefficients`	list	List of possible compositions for fixed-composition calculations or extremes for variable-composition simulations.	`List[PositiveInt]`		`[1 1]` `[2 1]`
Input	`maximum_number_of_components`	numb	The maximum number of components (atoms or	1-		4



https://github.com/COMCIFS/Structure_Prediction_Dictionary

CCDC

CSP Data Standard Paper

- The CSP data standards paper is currently being prepared.
- We would like the author list to reflect those who have contributed to the development of the dictionary. If you have provided feedback, suggestions, use cases, or participated in discussions that influenced the dictionary, please contact me (nfrancia@ccdc.cam.ac.uk).
- We welcome contributions to both the paper and the ongoing development of the dictionary but please get in touch by 10 December.
- All people in the meeting discussions will be acknowledged



A Data Standard for Crystal Structure Prediction Methods

Nicholas Francia[a], Lily M. Hunnisett[a], Jonas Nyman[a], Antanas Vaitkus[b], Andrius Merkys[b], Saulius Gražulis[b], Isaac J. Sugden[a], and Ian Bruno[a]

[a]The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, United Kingdom
[b]Section of Crystallography and Chemical Informatics, Institute of Biotechnology, Life Sciences Center, Vilnius University, Saulėtekio al. 7, Vilnius LT-10257, Lithuania

Acknowledgments

The CCDC CSP Data Standards Team

Nicholas Francia

Isaac Sugden

Lily Hunnisett

Jonas Nyman

Ian Bruno

CCDC staff

Contributors to discussions

Saulius Grazulis
Antanas Vaitkus
Hari Muddana
Dejan Zagorac
Jacco van der Streek
Kamil Dzuibek
Mike Bellucci
Shubham Sharma
Simon Westrip
Stefano Racioppi
Stephan Ruhl
Zhuocen Yang
Chris Pickard
Claire Adjiman
Andrius Merkys
Graeme Day
James Hester
Luca Ghiringhelli
Sally Price
Simon Coles
Simon Parsons
Zahra Momenzadeh
Erin Johnson
Arman Boromond
Susan Reutzel-Edens
Artem Oganov
Greg Price
Rui Guo
Jiri Klimes
Marc Meunier
Mihails Arhangelskis
Nikhar Rahul
Benjamin Tan
Jennie Martin

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